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How Much Life do I Lose from the Plague? Educational Board Games as Teaching Tools in Archaeology and Ancient History Courses

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This article describes the development, implementation, assessment, debriefing and outcomes, and potential uses of a university student-driven educational gaming project, as a major assessment item in an archaeology course entitled 'Ancient Medicine'. Discussion of these various aspects of the project is explored in the context of educational gaming theory and practice. We demonstrate that educational board games not only represent an effective method for students to retain and convey information in courses exploring the history and archaeology of medicine, but provide such a platform for undergraduate courses in general. Furthermore, we argue that they can present a cost-effective, fun and less time-/resource-intensive alternative to electronic or web-based projects, while still being an attractive addition to traditional classroom teaching methods.

KEYWORDS ancient history, archaeology, course design, educational board games, pedagogy

Introduction

During the Southern Hemisphere spring semester of 2010, the brand new course 'Ancient Medicine' was developed by one of us (MO) and delivered with the assistance of DH within the School of Archaeology and Anthropology, Australian National University. The content goal was to survey medical and surgical systems, in addition to medical theory and practices, from prehistoric times through to the early medieval period including: Greece, Rome, Egypt, Mesopotamia, India and

China (where historical and archaeological resources are most plentiful). Further, specific topics included contraception and abortion, development, transformation and transmission of medicine, and medicine and its instruments. An ancient medicine course, run within an archaeology department, facilitated a focus on the development and diversity of medical systems and practices from a ‘holistic’ perspective (incorporating often-overlooked archaeological data), thus obviating the need to focus solely on one specific historical period or medical system.

In designing an innovative course such as this, it was realized that a multi-component and group-based assessment project was needed to complement, but move beyond, student research conducted for their essay-based assignments. The course developer sought an approach that required students to learn and retain additional content in order to construct a ‘tool’ that they could then utilize to share with others the knowledge they had gained. Moreover, a device to collectively work through problems, and have fun while doing so, was considered an essential pedagogical outcome. It is our belief that these essential conditions can be met through student engagement in the development of a well-designed educational game, regardless of the format (web-based, digital, board games, card games, etc.).

The use of educational games in specifically archaeological or historical contexts is yet to realize its full potential, with those produced falling primarily within the category of simulations, modelling or virtual reality (i.e. the virtual reconstruction of an archaeological excavation). The difficult-to-achieve balance of fun, educational content, memorability and replay value is arguably still elusive (e.g. [Ardito, et al., 2007](#); [Burke & Smith, 2007](#); [Stone, et al., 2009](#)). Nevertheless, the complex nature of modern archaeological and historical research lends itself to being communicated to diverse audiences through gaming experiences. This is especially true if the students/players themselves are in charge of the design and testing process.

The aim of this paper is to describe the development, implementation, assessment, outcomes, and potential uses of this university student-driven educational gaming project. Before the project itself is described, a review of relevant educational gaming theory and contextual background is required to flag the importance of educational gaming with respect to our own specific purposes.

Why a learning game?

In his classic work *Homo ludens*, Dutch anthropologist John Huitzinga struggled with defining the relationship between concepts of play and seriousness, noting that one can ‘play seriously’, but that ‘seriousness does not seem to include the possibility of play’ ([Huitzinga, 1955](#): 3). This suggests that play supersedes it, and can thus ‘help constitute social and cultural functions of great gravity ... including religion, politics and warfare’ ([Huitzinga, 1955](#): 26–28). More recently, particularly since the 1970s, as computational, digital and online technologies flourished, a separate genre of serious games has emerged, seeking to incorporate the engaging elements of play (its memorability, interactivity and immersive nature) into discussion and education with respect to otherwise complex topics. [Abt \(1987: 9\)](#) defines them as games that have ‘an explicit and carefully thought-out educational

purpose and are not intended to be played primarily for amusement, but that this does not mean they are not, or should not, be entertaining’.

Although commercial games, in all their variety, are not necessarily incapable of simulating or illuminating real-world events, serious games tend to be principally identified through their approach to such topics. While institutions like the Serious Games Initiative – an association to promote the use of games for serious purposes (see [Djaouti, et al., 2011](#)) – tend to define such games as those designed to support the positions or campaigns of companies, institutions, politicians or advertising agencies, more recently those games that persuade players to both learn about and potentially challenge commonly held misconceptions about real-world systems are being included. These are termed *persuasive games* where one ‘makes arguments about the way systems work in the material world [in order to] alter or affect player opinion outside of the game ... to move the player from the game world into the material world’ ([Bogost, 2007](#): 47). These are games that make effective use of *process* to illustrate or simulate the complex. On a macro scale, [Sutton-Smith \(2001\)](#) defined seven rhetorics of play, such as progress and power, which define the broad cultural contexts that underlie the rules of games and what one intends to demonstrate with them. More relevant to discussions of the ‘power of play’ is a recognition that rhetoric and design can be used to ‘unpack the rules of a particular game in a particular context’ ([Bogost, 2007](#): 54).

A successful educational game will be purposefully designed to operate within contextualized situations relevant to its audience; situations actively participated in, but with outcomes either ignored or taken for granted. How can educational games accomplish this? Game theorist James [Gee \(2003\)](#) identifies a four-step process in which the player:

1. probes the virtual/simulated world depicted in the game through their actions;
2. reflects on the result and forms a hypothesis about its significance;
3. re-probes the world with that hypothesis in mind to examine its effect; and
4. treats that effect as feedback from the world in order to rethink their hypothesis.

After all, ‘a game is not simply a text to be read, but an experience to be had’ ([Consalvo, 2003](#): 173). Even for those games in which certain end-game scenarios can be viewed as negative (i.e. avatar death, disease infection), a player’s ability to actively manipulate their own outcome makes all such knowledge gained a worthwhile experience.

The theoretical grounding described above, and implemented in this project, originally derives from computer-mediated educational games, which themselves typically have long development time frames and substantial budgets. In the case of games designed to educate players about historic or archaeological topics or methodological approaches, appropriately innovative game mechanics have allowed players to explore historiographic traditions via integrated game mechanics ([Uricchio, 2005](#)), ‘rewind’ Australia’s colonial timeline in a platform game format ([Apperley, 2007](#)), or explore how the need to ‘master’ the intense game-play required in World War II first-person shooter games can provide a ‘hook’ for players to learn more about the key battles and events ([Fisher, 2011](#)). While computer games, and the task of ‘writing history through algorithms’, greatly expands the degree of

immersion and topical engagement that students can obtain (Graham, 2014), currently the best approach to electronically simulate broad historical trajectories or the minutiae of a specific event remains open to debate (Kee, et al., 2009). The complexities of digital game design, the need to incorporate training in basic coding (or request student familiarity with it as a prerequisite), and the more arduous debugging requiring to fix mechanical issues identified in play-testing, all make traditional board game design a viable alternative.

Scholars and game designers (e.g. Prensky, 2001; Ruggill & McAllister, 2008) continue to extol the virtues of well-designed learning games as relatively cheap, efficient ways to illustrate similarly complex subject matter, accommodate the differing attention spans and learning modes of diverse audiences, and even attract those with little to no previous gaming experience (such as children in developing countries), given appropriate delivery systems and context. Prensky (2001: 164) defines eight relatively binary categorizations useful for classifying a learning game. Through this classification the games designed by students in the course ‘Ancient Medicine’ can be seen as intrinsic (i.e. content is an integral part of game structure), tightly linked (i.e. constructed around specific information) and asynchronous (i.e. turn-based, allowing players to make decisions at their own pace, although occasionally within time limits). They are also session-based (i.e. exist only when a player is interacting with it; developments in the game cannot occur without the players causing them), and make efforts, each in their own way, to balance game play recollection and use of educational content through the narrative presented with player action requiring reflex. Overall, the *processual* nature of educational games was stressed in this project, so that even a non-electronic board game designed and constructed in a short time span, and with a shoe-string budget, could still provide accurate and memorable information about such complex topics as health, disease and medical systems throughout history.

Game board project goals and execution

From the outset, the project was conceived with several goals in mind that could be achieved over the course of the semester. As an initial step, during the first two weeks of the course students were asked to choose their teams (with ideal team numbers ranging from four to six), and initial tutorial sessions focused on getting to know each other and becoming familiar with what was expected from their final games in terms of educational value, practicality, replay value and ease of play. At this time we introduced the concept of educational games and how they differed, in terms of their use of rhetoric, text and graphical content, and player outcomes, from the commercial games that at least some students were familiar with. This initial foray into the topic was facilitated by the purchase and group trial of several ostensibly educational games with historical/archaeological themes, taking the time to dissect what did and did not work for each one.

With respect to the goal of enhancing educational content and student involvement in their own learning, the project allowed each team the maximum flexibility in developing the theme around which they built their game. Some chose to delve deeper into the medical system or practices of a particular time and place from

the perspective of patients, physicians, apprentices or society in general. Others took a comparative global approach, requiring players to gain and use knowledge about different medical systems while travelling around the ancient world. Still others placed the players in a more recent, but still archaeologically relevant, setting (for example, a 1930s expedition to Egypt in the style of Howard Carter), requiring that knowledge of ancient medicine be used to get out of danger. There was even a game created in which game play occurred at the individual and population level simultaneously, as players race to bring a cure to their chosen Medieval European city suffering from the plague. To communicate such complex information via a board game required extensive outside research in the form of academic books and journal articles as well as the Internet, to not only ‘get their facts straight’, but to modify the educational content presented as each game was developed and tested over the course of the semester.

Another initial step in the design process saw each team establishing roles in order to delegate responsibility for overseeing the key aspects of game design, such as outside research, artwork, game mechanics, quality assurance (i.e. the person in charge of responding to feedback from the ‘alpha’ and ‘beta’ testing sessions conducted half-way through the semester) and an overall project manager. In order for the students to better understand what each of these components of game design usually entailed, a crib sheet was created containing general descriptions of what was expected in each role. This quick reference guide was collated from numerous real-world commercial and educational examples, with particular reference to an ongoing learning game project with an educational focus on heritage preservation in the face of the rampant looting of antiquities in Cambodia (Huffer, 2009). Importantly, the adoption of such specific roles by team members was not mandatory, and certain tasks were shared or shifted as each group’s game developed and progressed.

During the first half of the semester classroom time was spent in brainstorming, initial research, creating prototypes of the game board and accessories required, and beginning to devise the rules of play. To facilitate this initial development a flow-chart was drawn up to illustrate how the various aspects of rhetoric and game mechanics might relate to each other within a well thought out educational game. This process again drew on real-world examples from pre-released games, and the trial and error involved in putting this into practice with an ongoing project. This was in preparation for a key required task; namely, alpha and beta play testing of each game, self-administered at first, followed by at least two rounds of exchange with another team. As recent literature stresses (e.g. Kelly, et al., 2007; Winn & Heeter, 2006), sufficient play testing is vital to the success of any game, educational or otherwise. Three weeks were devoted to such testing, which was facilitated by the creation of bug-catching spreadsheets on which play testers were asked to record any problems encountered in game play at that stage of development according to various categories (e.g. artwork, rules, mechanics, learning content, etc.), then offer suggestions on how to address each issue. Once each game had been through several testing iterations, with associated feedback, the remainder of the semester was devoted to finalizing their game to be ready for assessable play. Administering the project in this manner allowed the students to meet another aim, to better understand the difficult process of educational game design and what

distinguishes well or poorly constructed games, as well as develop their teamwork skills and ability to give constructive criticism.

Project assessment, outcomes and debriefing

Assessment centred around three specific criteria: (1) the overall success of the game itself; (2) individual and team member participation evaluations (used to track contribution to team effort and outcomes); and (3) a game overview document (used to moderate assessment of final game by assessing individual project input). For this final criterion students addressed the game's educational goal, intended audience, object, materials required, time limit, rules, design, and any remaining problems of note at the end of the semester. Each group was also requested to create a poster version of this document so that a visual representation (marketing tool) of each game could be showcased in subsequent semesters, as well as briefly display and describe (pitch) their finished game to the class before the final play-testing/evaluation session began. It is worth noting that the members of each team pooled resources to cover the costs of game construction, printing, etc., but when shared between groups of three to five individuals each, total costs amounted to substantially less than standard textbook fees.

The game itself was evaluated for its educational value (i.e. to win or effectively participate in the game, a player had to gain and retain knowledge of the ancient medicine theme(s) underpinning the story and rhetoric), its practicality (i.e. ease of set up and play within the intended time limit, and by the intended audiences), and the 'user-friendliness' of its rules (i.e. the rules are succinct, flow logically, and have clearly been tested, and adjusted as a result of earlier testing). As in the alpha and beta testing stages, each team received several opportunities to play the games of other teams within and outside of their tutorial session, and during each session the authors would join teams at random to evaluate game-play in action and ensure that each testing session remained on task. Evaluating the projects this way ensured that both individual and group participation was taken into account, and gave each team the chance to pitch their game to the fullest, in the manner that they might have to if approached by a board game production company. At the end of the semester, the members of each team were allowed to do what they would with the final product, including distributing copies of the game or even attempting to commercially market it. To our knowledge, no teams have attempted to produce their game commercially yet.

Overall, the authors were very impressed by the quality of the final games produced. Given that this project, and indeed the entire course, was a pilot and that only one of us (DH) had direct experience in educational game theory, design and testing, the fact that each team submitted a final, functional game with only an hour of contact time per week over the course of a four-month semester is commendable. Eleven final games were submitted: *Ancient Interns*, *Ankh – The Game of Life*, *Cursus Medicorum*, *Dyscracia*, *Favour of the Gods*, *Humourous Death*, *Medieval Mania (the game of Medical Madness)*, *Mummies and Maladies: The Curse of Imhotep*, *Sanctuary*, *Mundus Medicina*, and *Yin Yang: A Journey Through Chinese Medicine*. Student feedback was diverse, but overwhelmingly positive.

The distinctiveness and success of the project was even noted by the local university media as something that might be emulated (ANU, 2010).

To give two examples of games produced, we will discuss *Ankh – The Game of Life* and *Ying Yang: A Journey Through Chinese Medicine*. While all of the games produced were high-quality, the two selected were chosen due to their creative game play and markedly different subject matter. *Ankh – The Game of Life* was developed by a group of six students (Ash, Dean, Sang, Jasmine, Natalie and Jema), the chief educational outcome of which was to teach players about ancient Egyptian medicine (see Figure 1). To achieve this, players were required to navigate their way through a pyramid complex and exit the pyramid alive. Progression through the pyramid followed a throw of the dice, movement could be in any viable direction and choices were based on potential outcomes (four squares in one direction may land you on an illness square, whereas four moves in another direction may land you on an Ankh square and perhaps the chance to be resurrected by a ‘god’ card). Landing on illness squares required one to have a knowledge (or develop it quickly) of Egyptian medicine in order to avoid the illness. Other squares provided chance positive or negative outcomes. One, in particular, led to death and mummification, a condition that needed remedying prior to completing the game (you needed to be alive to win). In some ways this is a version of a general trivia game on a board replete with obstacles; there’s a die to facilitate movement, playing pieces, cards of chance and also question cards (the various illnesses one needed to overcome with medical knowledge). Students noted that the game was relatively easy to develop within the time frame available; however, issues such as clarifying and adjusting the rules, spell-checking question cards, as well as polishing



FIGURE 1 *Ankh – The Game of Life*. This is the final version of the game submitted for evaluation and grading.

the mechanics – for example, ensuring strategic elements actually worked (such as the ability to move in any viable direction to avoid or purposely land on specific squares) – were time-consuming.

Ying Yang: A Journey Through Ancient Chinese Medicine had the dual educational objective of instructing players on the history of Chinese medical systems as well as a range of Chinese medical treatments. Initially, five students were involved in this project with one dropping the course and one electing to complete a poster in lieu of the game, leaving Leisa, Sharyn and Claire to develop and finalize the game. This game also used player pieces, a die and a series of cards (see Figure 2). The pentagonal board included an outer and inner court, with a series of tasks that needed to be completed in the outer court before one could enter the inner. There was a significant use of Chinese symbolism and medical theory in the design of the game. For instance, a series of five element cards (wind, fire, etc.) needed to be collected during one's journey around the outer courtyard, while learning about this particular medical system and its history, before crossing a bridge into the inner courtyard. Here, element cards are exchanged for therapy disks, again in the context of learning about Chinese medicine and treatments. A stronger emphasis on learning, rather than game strategy, went into this very successful and elegant game.

These specific game examples aside, the success of this project did not come without challenges. For instance, as students were encouraged to choose their own teams it was often the case that friends teamed up, leaving other teams to be comprised of strangers (at the time). In both instances this sometimes led to personality conflicts which needed to be resolved in consultation with the teacher. The primary solution, when interpersonal conflicts could not be resolved, was to allow the student(s) concerned to instead create a conference-quality poster on an ancient medicine topic of their choice. Additionally, a web-based discussion forum

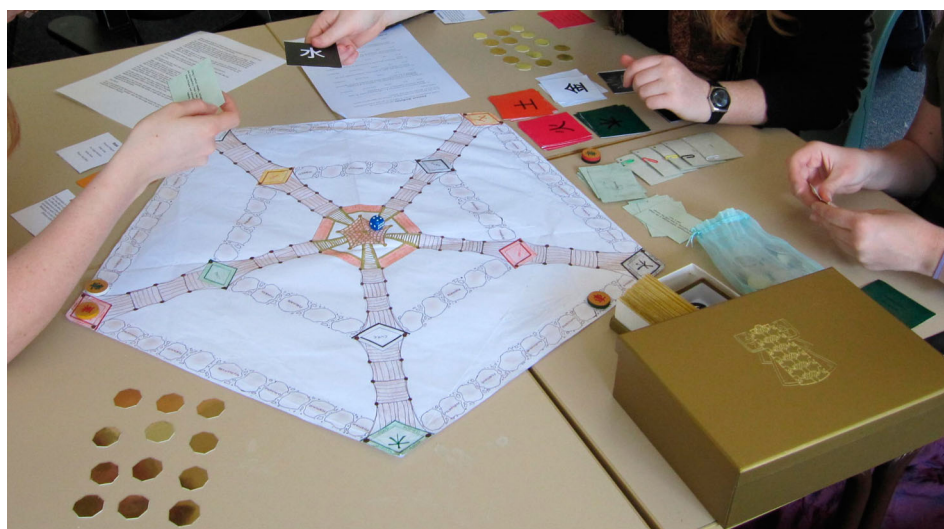


FIGURE 2 *Ying Yang: A Journey Through Ancient Chinese Medicine*. This is a draft (mock up) version of the game at the initial play testing stage.

was set up to provide the students a way to ask questions regarding game design, theory, etc., but unfortunately, this resource went almost entirely unutilized, most likely due to sufficient attention being paid each group in class as development progressed. Standard departmental class and teacher evaluation forms were filled out at the end of the semester by each student, in which the educational board gaming project achieved very positive feedback.

Overall, from the quality of the final games produced, it is clear that students took their tasks quite seriously and were getting enough out of the project to view it as much more than just another assignment. However, a few of the students offered private critique to the authors at the end of the semester to the effect that they personally felt that meeting our high expectations was detracting too much from their ability to complete all assignments for all classes. In regards to overcoming time constraint issues in constructing their games over the course of one semester, we feel that the scheduled structure of the semester was vital to their success. Each stage of the design process (e.g. preliminary designs, alpha testing and beta testing) was kept to strict time intervals of a few weeks, with teacher and students being able to participate in play-testing at each interval. Each round of play testing was assessable against various selection criteria, with each team required to provide evidence that play testing critiques were being actively responded to.

Notwithstanding, the time-constraint criticism was dealt with in subsequent iterations of this project for the 'Ancient Medicine' class, most recently offered in Southern Hemisphere spring semesters 2012 and 2014, by decreasing the number of formal play-testing sessions required and assessed, as well as allowing the individually created conference quality poster to be a valid form of assessment from the beginning. It is our opinion that, regarding the pilot phase of the project offered in 2010, by completing this project students were both able to view historical and archaeological research pertaining to ancient medicine in a different light, due to the hands-on approach requiring them to learn and then disseminate the material in game format, and also to begin to view gaming as potentially more than a hobby or purely commercial, electronic pastime.

Conclusion

This article described the creation and successful implementation of an educational board game project within a newly established course offered in the School of Anthropology and Archaeology, Australian National University, entitled 'Ancient Medicine'. Relevant theoretical background was addressed to contextualize our argument for the use of educational games, particularly board games, in the university classroom. It was demonstrated that an educational board game project would suit any course geared towards university students in which the topic covered is complex and multi-disciplinary, where content retention can be improved by group-based investigations, or where learning through hands-on student-driven projects is a desired outcome.

The vast majority of commercially released educational games are electronic, completed by professional teams on a large budget, usually over a multi-year time frame, or else specifically within a game design course in which the only requirement

is to learn about and build educational games (e.g. Chaffin & Barnes, 2010). As the role of the game project in ‘Ancient Medicine’ was to facilitate classroom-based learning (factual, critical and theoretical) about ancient medicine, other outcomes not normally associated with commercial game development were targeted. We believe we have demonstrated that educational board games present a cheap and less time-/resource-intensive alternative to electronic or web-based projects, while still being suitable to the objectives of both students and instructors. Budgeted expenses extended to craft supplies and time only. Without the necessity to cover the costs of specialized employee salaries or expensive computer hardware and software, educational board games are an attractive addition to traditional classroom learning in any economic region of the world, especially suitable for university departments on a budget or low socioeconomic status secondary or tertiary classrooms. The results of such projects can extend beyond education, with finished products becoming symbols of pride, and with some even being potentially marketable.

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